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Studierichting: Informatica

Oefeningenreeks 1B nr. 20.6

Bereken het volgende getal in $\mathbb{C}$:

$$\begin{aligned} & \frac{1}{(1+i)^3} + \frac{1}{(1-i)^3} \\ &= \frac{1}{(1+i)^2(1+i)} + \frac{1}{(1-i)^2(1-i)} \\ &= \frac{1}{2i(1+i)} + \frac{1}{-2i(1-i)} \\ &= \frac{1}{-2+2i} + \frac{1}{-2-2i} \\ &= \frac{-2-2i}{(-2+2i)(-2-2i)} + \frac{-2+2i}{(-2-2i)(-2+2i)} \\ &= \frac{-2-2i}{8} + \frac{-2+2i}{8} \\ &= \frac{-4}{8} \\ &= -\frac{1}{2} \end{aligned}$$

References